



MCT5210 – Control Engineering III

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Course Outline

1. State Space Analysis

State space description of linear systems

Linearisation of Nonlinear Systems

Relationship between state equations and transfer functions.

2. Controllability and Observability

Concepts of controllability and observability

Observer design

3. Principles of Sampled Systems

Z-transforms

Stability in the Z-plane

Derivation of Digital Control algorithms

Microprocessor application



Course Outline

4. Introduction to Nonlinear Systems

Types of Nonlinearities

Describing Functions

Lyapunov analysis

Stability of Nonlinear systems

Introduction to adaptive control



Assessment

- Group Assignment/presentation (20)
- Test (20)
- Exam (60)



References

- Kuo, Benjamin C., and Farid Golnaraghi. *Automatic control systems*. 9th Edition. Englewood Cliffs, NJ: Prentice-Hall, 2009
- Ogata, Katsuhiko. *Modern control engineering*. Vol. 5. Upper Saddle River, NJ: Prentice hall, 2010.
- Ding, Z. *Nonlinear and Adaptive Control Systems*. London, UK: Institution of Engineering and Technology. 2013.